Project plan for a team of four

11 October to 2 November · Track 2, False Data Injection · submit Monday 2 November (deadline Tuesday 3 November, 23:59 GMT+1)

Member	Owns	Rubric lines carried	Report sections and figures
1 **Data, Sentinel-2 and integration**	`data`, Sentinel-2 context, `scripts`, the repository, report assembly, submission	Feasibility and local execution (10) Clarity and reproducibility (5)	Datasets, feasibility, README. Figure 2.
2 **Signal processing, response and product**	`processing`, `response`, `product`, the with/without comparison, the ablation	Signal processing (15) Part of standard scenario and evasion	Signal processing, response policy. Figures 3, 4, 10, 12.
3 **Trust, attacks and threat model**	`trust`, `sensor`, `transport`, `attacks`, the evasion library	Threat model and relevance (15) Architecture and trust mechanism (15)	Threat model, architecture. Figure 1.
4 **Detection and evaluation**	`detect`, `eval`, every result table	Detector design and evaluation (20) Standard scenario and evasion (20)	Detectors, evaluation, results. Figures 5 to 9, 11.

The whole project at a glance

Member	Now to Mon 19 Oct The standard scenario gives numbers	Tue 20 to Mon 26 Oct Context, learning, evasion	Tue 27 Oct to Mon 2 Nov Harden and write
1 **Data**	Repository skeleton, NASA POWER client, sites and station loaders, file mode	Sentinel-2 context, `run_all`, dataset table	Blind rehearsal 2, report assembly, clean export, submission
2 **Signal**	Quality control, alignment, calibration, residuals, validation gate	Decision product, the four responses, with/without comparison	Ablation, fresh-clone test on a second laptop, figures
3 **Trust**	The three official injections, trust module, threat-model draft	Stream mode at the three attacker positions, evasion library, overhead, blind rehearsal 1	Remaining injections, attack-to-control mapping, residual risk
4 **Detection**	Detector S, scenario runner and metrics, first standard-scenario table	Detector M, probability and trust score, frozen thresholds, held-out sites, evasion runs	Full runs, confidence intervals, final figures
Team gate	**Mon 19: one command runs the standard scenario on the development sites and prints its numbers**	**Mon 26: every result table is filled, even if rough**	**Thu 29 report complete · Fri 30 review · Sun 1 Nov freeze · Mon 2 Nov submit**

Rules for all four

- **Monday 12 October, before anyone codes:** agree the reading table schema, the residual vector layout, the injection label format and the config keys.
- Write your report section in the week the work is done, not in the last two days.
- No names, university or branch anywhere in the repository or the report.

Where the load sits

- Member 4 carries 40 of the 100 points. Member 2 takes the with/without comparison and the ablation to share that load.
- Member 3 has the heaviest single week (20 to 26 October).
- If time runs short, follow the priority order in `docs/01`: items 1 to 6 are the submission.

1 Member 1: Data, Sentinel-2 and integration

OWNS
`data`, Sentinel-2 context, `scripts`, the repository, report assembly

RUBRIC
Feasibility and local execution (10), clarity and reproducibility (5)

DEPENDS ON
Nobody at the start. Everyone depends on this role.

	Now to Mon 19 Oct The standard scenario gives numbers	**Tue 20 to Mon 26 Oct** Context, learning, evasion	**Tue 27 Oct to Mon 2 Nov** Harden and write
What to do	1. Send the 8 organizer questions today. 2. Repository skeleton, pinned `requirements.txt`, pytest, neutral git identity (Mon 12). 3. NASA POWER client with on-disk cache and checksums **(due Wed 14)**. 4. Pick 6 to 8 development sites and 2 held-out sites; write the station loaders. 5. Label extreme-weather and provider-fault periods. 6. Register the Copernicus account. 7. `scripts/assess` file mode end to end, wiring in the modules of Members 2, 3 and 4.	1. Sentinel-2 context extraction: small windows around each site, reduced to cloud, shadow and snow fraction, NDVI, NDMI and NDWI. Store only the indicators and the product identifiers. 2. Build the orbital-context residual with Member 2: sky, water-balance and snow agreement. 3. Dataset table: provider, licence, sites, date range, product identifiers, checksums. 4. `scripts/run_all` with a quick mode (one site, a few minutes) and a full mode; ship the small cache so it runs offline. 5. Draft the feasibility section: hardware, run time, zero-cost bill of materials.	1. Blind rehearsal 2: inject the three attacks at a new site and interval; hand over only the series and the coordinates. 2. README and documentation pass. 3. Assemble the report from the four members' sections: complete Thu 29, review Fri 30 (you review Member 2's sections). 4. Export to a fresh repository with a clean history; run the anonymity checklist; clear the PDF metadata. 5. Freeze Sun 1 Nov. Submit Mon 2 Nov.
Finished when	One command with latitude, longitude and dates returns the station and reference tables in the agreed schema for every site, and a second run works offline. On a fresh clone, one command runs the standard scenario on the development sites and prints detection rate, latency and false alarms.	A context table exists for every site, built from the cache. `run_all` regenerates every draft table from one command.	Every box of "Definition of done" in `docs/06` is ticked. The submission is sent on Mon 2 Nov.
Main problem	Station archives are messy: different units, fill values and day definitions, with real faults inside "clean" data. Everyone else is blocked until the data lands.	Sentinel-2 access can stall on registration, formats and cloud cover. Against a 12 % step it adds little, so the pipeline must run without it and the report must say so honestly.	The working git history carries a real author name on every commit. Submitting it as it is risks disqualification, so the clean export is not optional.

2 Member 2: Signal processing, response and product

OWNS
`processing`, `response`, `product`, the with/without comparison, the ablation

RUBRIC
Signal processing (15), part of standard scenario and evasion

DEPENDS ON
Member 1's data and Member 3's injections, both by Wed 14

	Now to Mon 19 Oct The standard scenario gives numbers	**Tue 20 to Mon 26 Oct** Context, learning, evasion	**Tue 27 Oct to Mon 2 Nov** Harden and write
What to do	1. Quality-control tests: gross range, seasonal range, spike, rate of change, flat line, internal order, timing. 2. Day-boundary alignment (lag of -1, 0 or +1 day). Gaps are marked, never interpolated. 3. Robust per-season calibration against the reference, in calibrated and cold-start modes. 4. Residuals: temporal, physical, reference (difference and log-ratio), replay features. 5. Validation-gate plots on clean data.	1. Add the orbital-context residual with Member 1. 2. `product`: a 30-day water balance (rain minus FAO-56 reference evapotranspiration) mapped to a few decision levels. 3. `response`: de-weighting, verification request, quarantine with substitution from the calibrated reference, alert. Silent stations are excluded. 4. With/without comparison: wrong-decision rate for configurations C0 to C4, using Member 4's runner. 5. Draft the signal-processing section with figures 3 and 4.	1. Ablation table: which residual groups matter, including what Sentinel-2 adds. 2. Fresh-clone reproduction test on a second laptop (you did not write the scripts) **(by Thu 29)**. 3. Time and memory per reading for quality control and residual generation. 4. Final figures 3, 4, 10 and 12. 5. Review Member 4's sections on Fri 30.
Finished when	The validation gate passes: on clean data at every development site, each residual is centred near zero, no larger in extreme weather than on ordinary days, and similar across climates. The plots are saved.	In at least one reproducible scenario, a 12 % step changes the decision level undefended (C0) and does not when defended (C4). The headline decision table in `docs/05` has draft numbers.	The ablation table is final. The headline table reproduces on the second laptop from a fresh clone with one command.
Main problem	A 12 % step is about the size of the normal station-to-grid difference. If calibration is loose the attack hides in it, and if residuals grow in heatwaves they cause false alarms.	A 12 % step may not move a 30-day water-balance level at all, and then the comparison shows nothing. Use sites and periods near a level boundary, and state that choice openly.	Reproduction usually breaks on unpinned versions, absolute paths or a missing cache. Run the test by Thursday so there is time to fix what it finds.

3 Member 3: Trust, attacks and threat model

OWNS
`trust`, `sensor`, `transport`, `attacks`, the evasion library

RUBRIC
Threat model and relevance (15), architecture and trust mechanism (15)

DEPENDS ON
Member 1's data by Wed 14; the organizers' answer on what "12 %" means

	Now to Mon 19 Oct The standard scenario gives numbers	**Tue 20 to Mon 26 Oct** Context, learning, evasion	**Tue 27 Oct to Mon 2 Nov** Harden and write
What to do	1. The three official injections with exact per-day labels and one YAML config each: 12 % step, drift to 15 % over 30, 60 and 90 days, 7-day replay **(due Wed 14)**. 2. Each one also under the three meanings of "12 %": of the value, of the range, of the standard deviation. 3. Trust module: Ed25519 keys, enrolment with position, sign, verify, sequence, hash chain, with unit tests. 4. Draft the threat-model section of the report.	1. `sensor` and `transport` emulation; stream mode at the three attacker positions (station, telemetry, pipeline). 2. Tamper-evident ledger, cadence watchdog, revocation. 3. Evasion library E1 to E9 as YAML scenarios; the white-box one with Member 4. 4. Measure the overhead: bytes per record, sign time, verify time. 5. Blind rehearsal 1: inject at a site and interval the others do not know; hand over only the series and the coordinates. 6. Draft the architecture section: block diagram, sequence diagram, record format.	1. Remaining injection suite (S4 to S11) and the held-out injection type. 2. Attack-to-control mapping: every evaluated injection is in the threat model; every threat is evaluated or declared out of scope with a reason. 3. Residual-risk statement: where the defence stops working. 4. Final threat-model and architecture sections by Thu 29, with figure 1. 5. Review Member 1's sections on Fri 30.
Finished when	One command reproduces the three injected series and labels for 3 sites, identical for the same seed. Unit tests pass for forged, modified, replayed and missing records.	In stream mode, injections on the telemetry and in the pipeline are rejected by verification, and the one at the station passes it and is left to the detectors. Every evasion runs from a config with exact labels. The overhead is measured.	The mapping table has no row without a control or a stated reason. The trust unit tests pass on the fresh clone.
Main problem	Nobody knows yet what "12 %" is relative to, or which series the jury attacks. Every reading must stay runnable from config.	This is the heaviest week of the four roles, and evasion work has no natural end. The floor is one experiment per defence layer. Finish that before adding more.	The remaining injections are priority 7 of 8: drop scenarios before they delay the report. Defences that failed must be written down, not hidden.

4 Member 4: Detection and evaluation

OWNS
`detect`, `eval`, every result table

RUBRIC
Detector design and evaluation (20), standard scenario and evasion (20)

DEPENDS ON
Member 2's residuals (Thu 15, Fri 16) and Member 3's injections and evasions

	Now to Mon 19 Oct The standard scenario gives numbers	**Tue 20 to Mon 26 Oct** Context, learning, evasion	**Tue 27 Oct to Mon 2 Nov** Harden and write
What to do	1. Start Mon 12 on fake residuals in the agreed layout; do not wait for Member 2. 2. Detector S: running-sum test, change-point search with `ruptures`, trend test, replay matcher. 3. Scenario runner and metrics: episode detection rate, latency, onset error, false alarms per station-year. 4. Replace the expected-difficulty table in `docs/04` with measured numbers.	1. Detector M: benchmark Isolation Forest, ECOD, COPOD, HBOS and a nearest-neighbour method from PyOD on clean windows; pick one. 2. A supervised random forest as an upper-bound reference only. 3. Fusion to a calibrated compromise probability; station trust score and trust bands. 4. Choose thresholds on validation data, then freeze them. 5. Held-out sites; run blind rehearsal 1 once, without tuning. 6. Run the evasion suite; compute the stealth budget per variable. 7. Latency and memory of the detectors; draft the detector section.	1. Full runs with final seeds, starting Tue 27; block-bootstrap confidence intervals. 2. Held-out injection result; blind rehearsal 2, once, without tuning. 3. Final figures 5 to 9 and 11. 4. Evaluation section by Thu 29, stating the number of readings, episodes and sites behind every figure. 5. Review Member 3's sections on Fri 30.
Finished when	The standard-scenario table for development sites and the per-variable table in `docs/05` hold real numbers: episodes detected, median latency, onset error, false alarms per station-year.	Every result table in `docs/05` is filled, even if rough. The thresholds are frozen and committed before the test period is touched.	Every number in the report is regenerated by `run_all` and carries a confidence interval. The figures are final.
Main problem	This role is last in the chain, so the real residuals arrive late. Wind and rain will probably not be detectable at 12 %, and that must be reported, not hidden.	Leakage: tuning anything after seeing the test period, the held-out sites or the rehearsal result makes every number worthless. Split by time and by site only.	Full runs are slow, and a bug found on Wednesday means running everything again. Use quick mode to catch bugs before each full run.